

Chi-square test $\rightarrow$ Categorical data

2010 $\rightarrow$ Weights of individuals in a small city &

want to compare
expected & observed
values

< 50 kg	50 - 75 kg	> 75 kg
20%	30%	50%

Population data

Expected

< 50 kg	50 - 75 kg	> 75 kg
$20\% \times 500$ 100	$30\% \times 500$ 150	$50\% \times 500$ 250

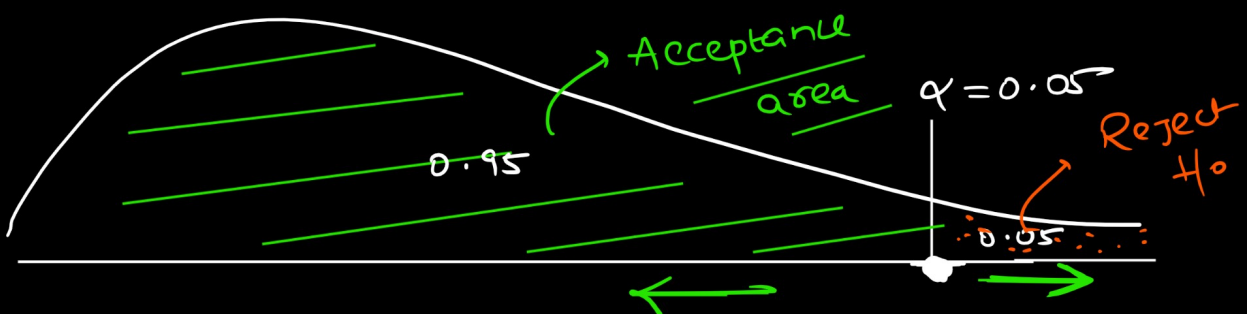
$n = 500$

2020

< 50 kg	50 - 75 kg	> 75 kg
140	160	200

$\rightarrow$ observed

Whether the population diff of weights have been changed
in the last 10 years



① $H_0 \rightarrow \text{Observed} \approx \text{Expected}$
 $H_1 \rightarrow \text{Observed} \neq \text{Expected}$

② $\alpha = 0.05$

③

$$\chi^2 \approx \frac{\sum (\text{Observed} - \text{Expected})^2}{\text{Expected}}$$

$$\frac{(140 - 100)^2}{100} + \frac{(160 - 150)^2}{150} + \frac{(200 - 250)^2}{250}$$

$\rightarrow$ for all the possible categories

$$\chi^2 \approx 26.66$$

Degree of freedom
 $\rightarrow 2$

④

$$p\text{-value} = 0.0001 < \alpha$$

$\rightarrow$ Reject H_0

Imagine you're a data analyst working for a retail company. The company wants to know if customer preferences for different types of products are evenly distributed across different age groups. To investigate this, you conduct a survey and collect the following data:

Product	Under 30	30-50	Over 50	Total
Electronics	50	30	20	100
Clothing	30	40	30	100
Home Goods	20	30	50	100
Total	100	100	100	300

→ observed
data

Question: Is there a significant association between age group and product preference?

Expected data

Product	age		
	← Under 30	30-50	→ > 50
Electronics	$100/3 \approx 33.33$	$100/3$	$100/3$
Clothing	$100/3 \approx 33.33$	$100/3$	$100/3$
Home Goods	$100/3$	$100/3$	$100/3$

evenly distribution

① $H_0 \rightarrow$ observed $\approx$ expected

← $H_1 \rightarrow$ observed $\neq$ expected

② $\alpha = 0.05$

③ $\chi^2 = \sum (\text{observed} - \text{expected})^2$

Expected

$$\approx \frac{(50 - 33.33)^2}{33.33} + \dots$$

$$\chi^2 = \underline{30.0}$$

$$df \approx (row - 1) * (col - 1)$$

$$\approx (3 - 1) * (3 - 1)$$

$$\approx 4 \checkmark$$

ANOVA Test $\rightarrow$ compare the mean
of three or more
groups

- ① factors
- ② levels

	<u>levels</u>		
<u>Medication</u> $\rightarrow$	5mg	10mg	15mg
<u>factor</u>			

Assumptions of ANOVA

CLT

- $\hookrightarrow$ Normality of sampling distribution of mean
- $\hookrightarrow$ Absence of outliers
- $\hookrightarrow$ Homogeneity of Variance
 $\sigma_1^2 = \sigma_2^2 = \sigma_3^2$

↳ samples are independent & random

Type of ANOVA

① one way ANOVA { ↳ 1 factor with atleast 2 levels & these levels are independent

<u>Medication</u>	5mg	10mg	15mg

② Repeated Measures ANOVA → one factor with atleast 2 levels & these levels are dependent

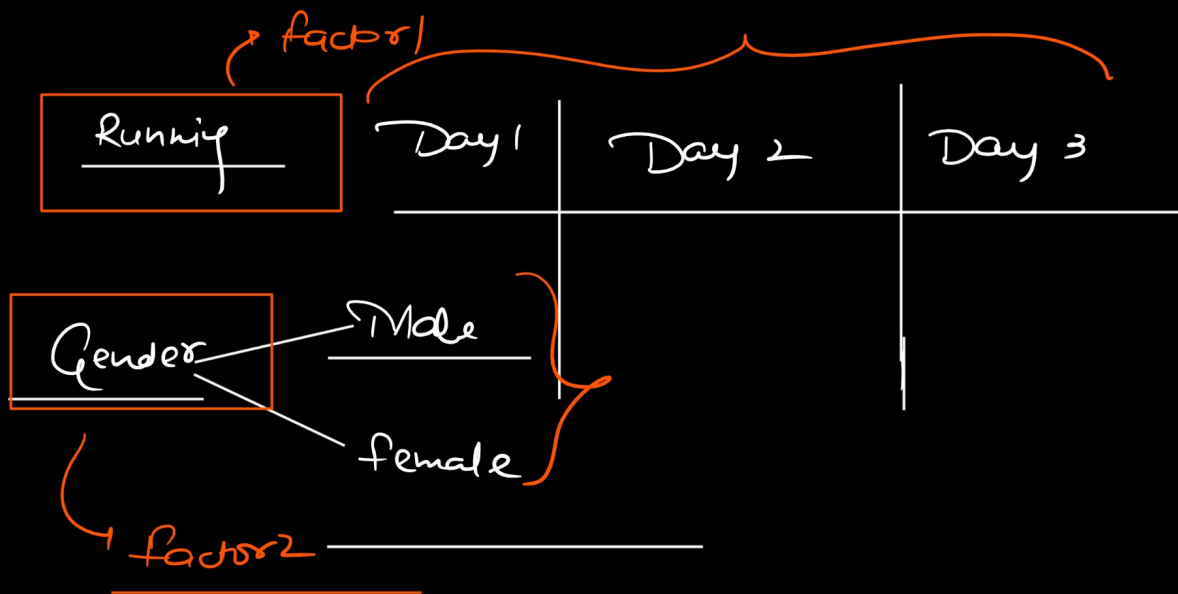
↳ factor
Running

Level → dependent		
Day 1	Day 2	Day 3

factorial ANOVA

↳ two or more factors each of
these factors have atleast
two levels

↳ independent or
dependent



$$F = \frac{\text{MSB}}{\text{MSW}}$$

MSB (Mean Square Between) is the numerator, representing the variation between groups.

MSW (Mean Square Within) is the denominator, representing the variation within groups.

equivalent → believe (standard)

$H_0 \rightarrow \mu_A = \mu_B = \mu_C$
 (Vice-versa)
 $H_1 \rightarrow$ there is a difference b/w μ_A, μ_B & μ_C

Teaching Method	^A Virtual	^B Physical	^C Hybrid
	8	6	9
	7	5	8
	9	7	10
	$\mu_A = 8$	$\mu_B = 6$	$\mu_C = 9$

$$\frac{\mu_A + \mu_B + \mu_C}{3}$$

$$\text{overall mean} \Rightarrow \frac{8 + 6 + 9}{3} \Rightarrow 7.67$$

$$f = \frac{\text{b/w group variance}}{\text{within group variance}}$$

num of data points in each group $\rightarrow$ group mean $\rightarrow$ overall mean

$$\text{b/w group variation} \rightarrow 3 \times \left\{ (8 - 7.67)^2 + (6 - 7.67)^2 + (9 - 7.67)^2 \right\}$$

$$\text{SSB (sum of square b/w the group)} \Rightarrow 14$$

$$\text{Within group Variation} \rightarrow \text{A spread} \rightarrow (8-8)^2 + (7-8)^2 + (9-8)^2 = 2$$

$$\text{B spread} \rightarrow (6-6)^2 + (5-6)^2 + (7-6)^2 = 2$$

$$\text{C spread} \rightarrow (9-9)^2 + (8-9)^2 + (10-9)^2 = 2$$

$$SSW = 2 + 2 + 2 = 6$$

$$f\text{-value} = \frac{MSB}{MSW} = \frac{7}{1} = 7$$

$\rightarrow$ A, B, C
 num of groups
 $k-1$

$$MSB = \frac{SSB}{df_b} = \frac{14}{2} = 7$$

$$MSW = \frac{SSW}{df_w} = \frac{6}{6} = 1$$

$$\frac{\text{num of observations}}{n-k} = \frac{9-3}{6} = 6$$

$\rightarrow$ num of groups

$$p\text{-value} \rightarrow 0.0270 < \alpha(0.05)$$

$\rightarrow$ reject H_0